

AlchemBERT: Exploring Lightweight Language Models for Materials Informatics

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Abstract

The emergence of large language models (LLMs) has spurred numerous applications across various domains, including material design. In this field, an increasing number of generative models focus on directly generating materials with desired properties more accurately, moving away from the traditional approach of enumerating vast numbers of candidates and relying on computationally intensive screening algorithms. However, we assert that without accurate prediction capabilities, effective material design is unattainable, as generating necessary structures becomes futile if their quality cannot be reliably evaluated. Matbench provides an excellent foundation for predictive tasks, yet prior efforts with LLMs have primarily focused on composition-related tasks using models such as GPT or LLaMA. In this study, we revisit BERT, a relatively small LLM with 110 million parameters, which is significantly smaller than GPT or LLaMA models containing billions of parameters. Remarkably, we demonstrate that BERT-base achieves comparable performance to these larger models in material property prediction tasks. Extending beyond composition tasks, we introduce BERT's application in structure prediction using CIF (Crystallographic Information File) data and natural language descriptions of structures. Our results rival state-of-the-art composition models such as CrabNet and, in several tasks across datasets ranging from a few hundred to over a hundred thousand samples, even surpass traditional structure-based and knowledge-driven models. Additionally, on the latest Matbench test task, Matbench-Discovery, our model outperformed the Voronoi RF model and achieved MAE results comparable to other models that rely solely on energy predictions. Our findings provide a new reference point for future LLM applications in material design, offering valuable insights for leveraging language model in this domain and emphasizing natural language descriptions over conventional model-centric design. We term this

application of BERT in material design AlchemBERT, signifying its novel role in bridging natural language and structural representations.

Keywords

materials property prediction, machine learning, language models, BERT

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Abstract

The emergence of large language models (LLMs) has spurred numerous applications across various domains, including material design. In this field, an increasing number of generative models focus on directly generating materials with desired properties more accurately, moving away from the traditional approach of enumerating vast numbers of candidates and relying on computationally intensive screening algorithms. However, we assert that without accurate prediction capabilities, effective material design is unattainable, as generating necessary structures becomes futile if their quality cannot be reliably evaluated. Matbench provides an excellent foundation for predictive tasks, yet prior efforts with LLMs have primarily focused on composition-related tasks using models such as GPT or LLaMA. In this study, we revisit BERT, a relatively small LLM with 110 million parameters, which is significantly smaller than GPT or LLaMA models containing billions of parameters. Remarkably, we demonstrate that BERT-base achieves comparable performance to these larger models in material property prediction tasks. Extending beyond composition tasks, we introduce BERT's application in structure prediction using CIF (Crystallographic Information File) data and natural language descriptions of structures. Our results rival state-of-the-art composition models such as CrabNet and, in several tasks across datasets ranging from a few hundred to over a hundred thousand samples, even surpass traditional structure-based and knowledge-driven models. Additionally, on the latest Matbench test task, Matbench-Discovery, our model outperformed the Voronoi RF model and achieved MAE results comparable to other models that rely solely on energy predictions. Our findings provide a new reference point for future LLM applications in material design, offering valuable insights for leveraging language model in this domain and emphasizing natural language descriptions over conventional model-centric design. We term this application of BERT in material design AlchemBERT, signifying its novel role in bridging natural language and structural representations.

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Introduction

The appearance of ChatGPT¹ has taken us into a new era. Large language model (LLM) characterized by billions of parameters and trained on extensive textual corpora, has redefined the boundaries of traditional language models, which typically possess only millions of parameters. LLMs are designed to understand, generate, and manipulate natural language with unprecedented accuracy and contextual depth, driving advancements in diverse fields.²⁻⁴ There is a growing belief that LLMs encapsulate the complexities of the real world within their parameters, enabling them to perform a myriad of tasks with unprecedented efficacy.⁵

In the field of materials science, high-throughput screening has been a prominent topic for several decades, with numerous frameworks and algorithms developed over the years.⁶⁻⁹ Typically, the approach involves an enumeration phase followed by a screening process, a strategy that can also be applied to LLM. To integrate material structures into an LLM, it is essential to develop a method for converting the structure into a sequence, ideally in a manner that allows for accurate reconstruction. A widely adopted approach is the use of the Crystallographic Information File (CIF),¹⁰ a standardized format for describing crystal structures developed by the International Union of Crystallography. Although we demonstrate that CIF is not the optimal method for regressing properties, this descriptor remains so fundamental that both LLaMA and ChatGPT can comprehend the CIF format and interact effectively with users without requiring any fine-tuning. We attribute this capability to the widespread inclusion of CIF data in pretraining datasets. A study by Antunes et al.¹¹ demonstrated the use of GPT to interact with CIF files for generating stable materials. The findings of this study were made publicly accessible through the website <https://crystallm.com>.

In the search for the best material candidates, some research efforts aim to generate a latent space that is systematically related to specific material properties, enabling more precise sampling based on the target property. Notable work in this area includes studies

by Gómez-Bombarelli et al.¹² and Xie et al.¹³. A similar approach has been applied to LLMs in material design. Recent studies on generative approaches for stable materials have shown that LLMs can be effectively utilized in material discovery. For instance, Park et al.¹⁴ introduced Chemeleon, a generative model based on a BERT-based¹⁵ variant (Crystal CLIP) that uses contrastive learning to align textual descriptions with 3D crystal structures. By incorporating diffusion models, it can efficiently generate stable materials within complex chemical spaces such as Li-P-S-Cl, highlighting the potential of generative AI in material design. Furthermore, Meta has explored the use of the open-source LLM, LLaMA2, for identifying stable materials.¹⁶ Their results demonstrated that LLaMA2 outperforms the Crystal Diffusion Variational Autoencoder (CDVAE),¹³ achieving twice the effectiveness in material prediction.

Throughout the history of materials science, numerous methods have been developed to evaluate materials by testing various properties, with a primary focus on assessing their stability and suitability for specific applications. Consequently, there is a strong correlation between a material's properties and its structural features.¹⁷ If a latent space truly exists that captures the tendencies of a specific property and can be used to sample materials, it must also be capable of regressing that property. Otherwise, its success in material generation could merely be attributed to chance. Therefore, evaluating whether an LLM can accurately regress material properties is essential to validate its effectiveness.

Matbench¹⁸ provides an excellent foundation for predictive tasks in materials science. Researchers have evaluated the performance of LLMs on the compositional tasks within Matbench. Although models such as GPTChem¹⁹ and Darwin²⁰ have shown promising results—significantly outperforming Dummy model—their evaluations are limited to composition tasks based on formulas, with no inclusion of structural tasks. Furthermore, as of now, no LLM has been tested on Matbench-Discovery.^{21,22}

In this work, we aim to evaluate the capability of LLMs to predict material properties, extending beyond composition-related tasks to include structural tasks and Matbench-

Discovery, which, to our knowledge, have not been previously tested with LLMs. In addition to using the CIF format, we also explore the use of natural language descriptions for crystal structures. For example, we might say **mp-13** in Material Project²³ as: **The structure’s cell parameters are (2.86Å, 2.86Å, 2.86Å, 90°, 90°, 90°), the space group is 229 (Im-3m), and a Fe atom occupies its 2a Wyckoff position, (further details are omitted here for brevity).**

More importantly, although BERT¹⁵ and its variants have demonstrated exceptional performance on language benchmarks such as GLUE (General Language Understanding Evaluation)²⁴ and SQuAD (Stanford Question Answering Dataset),²⁵ their application to Matbench has been largely overlooked. In this work, we also address this gap by providing a baseline for lightweight language models. Our findings reveal that lightweight language models outperform many human-designed models and perform on par with larger models, such as GPT, which contain hundreds of billions of parameters. As a result, this research suggests a future where physicists might simply describe a structure in natural language, eliminating the need for designing intricate models or laboriously screening material features.

Results

Since the pretraining of language models typically involves large amounts of data, it is crucial to ensure that no data leakage occurs with the matbench dataset. The results of this verification are presented in Fig. 1, which shows the performance of BERT without training on matbench data. All material property predictions are at the level of a Dummy model, regardless of whether the input is raw (e.g., CIF files for structure tasks or chemical formulas for composition tasks in matbench) or provided as natural language descriptions. Among all tasks, the results for four tasks (log_gvrh, log_kvrrh, mp_e_form, and perovskites) are worse than those of the Dummy model. For the remaining tasks, the performance of the model is closer to the Dummy model, approximately at the 20% mark on the scale between Dummy

and Best. This confirmation of no data leakage ensures that our subsequent discussion is meaningful.

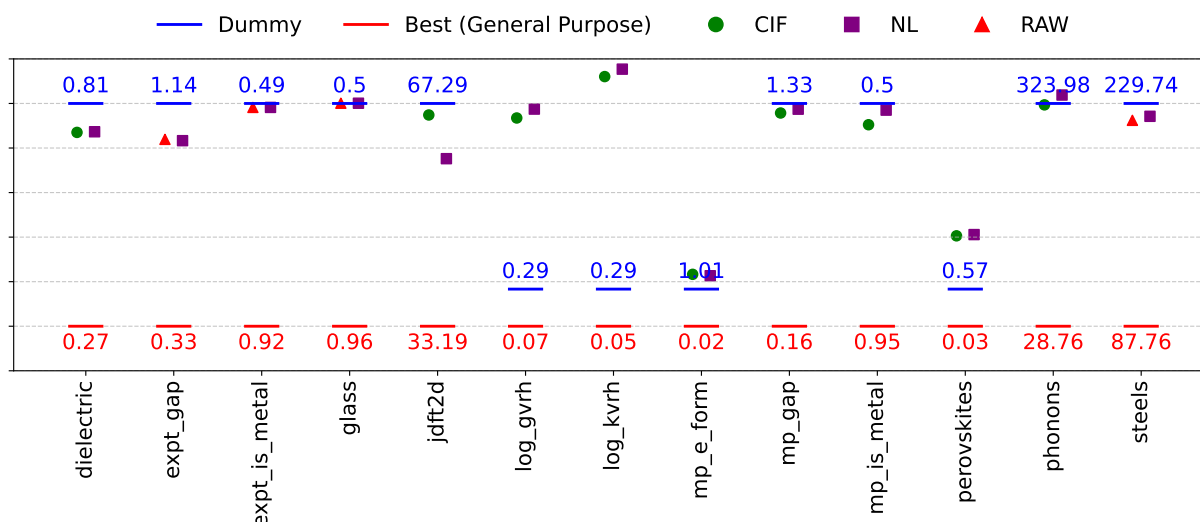


Figure 1: The matbench performance of BERT without training. The values beside the “Dummy” and “Best” lines represent the reported metrics for each task. “CIF” and “RAW” refer to the original formats used in matbench for structure and composition tasks, respectively, while “NL” represents the transformed natural language format generated by us.

We then followed the official training and testing procedure of Matbench to evaluate the different input formats, as illustrated in Fig. 2. The overall performance on structural tasks (using both CIF and NL inputs) is better compared to composition tasks, with all results achieving over 80% of the best values. The results demonstrate that the natural language transformation is effective, with an average improvement of 40.3%. Therefore, we use the NL input results to compare with other models on the matbench leaderboard, as shown in Fig. 3, and submit our model under the name AlchemBERT.

In Fig. 3, we compare our AlchemBERT model against CrabNet²⁶ and GPTChem.¹⁹ GPTChem¹⁹ is based on GPT-3,²⁷ a language model with 175 billion parameters—roughly 1,000 times larger than BERT.¹⁵ Despite this significant difference in model size, the performance on the Matbench composition tasks does not show a substantial gap. CrabNet,²⁶ a manually designed model, also does not demonstrate a significant advantage over Alchem-

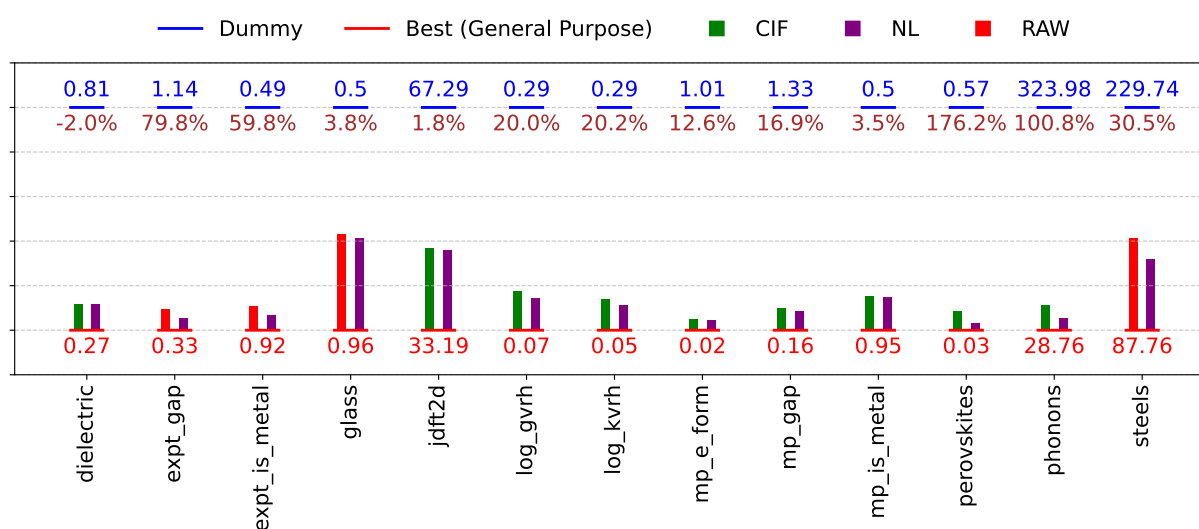


Figure 2: Results from the trained models. Shorter bars indicate better performance. “CIF” and “RAW” refer to the original formats used in matbench for structure and composition tasks, respectively, while “NL” represents the transformed natural language format generated by us. The percentile values in the figure represent the performance differences for each task, with positive values indicating that the natural language (NL) input outperforms the raw input.

BERT. Notably, on the “perovskites” task, which incorporates structural details, language models such as AlchemBERT outperform CrabNet²⁶ significantly. This is likely because CrabNet²⁶ relies solely on compositional information. However, incorporating even basic structural information can lead to significant improvements, as all the data pertain to perovskite-like structures. On certain specific tasks (See the Matbench¹⁸ leaderboard for details), AlchemBERT even surpass traditional structure-based models like CGCNN,^{18,28} Deep-erGATGNN,²⁹ MegNet,³⁰ DimeNet++,^{31,32} and knowledge-driven models such as MOD-Net.^{33,34}

These results suggest that while AlchemBERT may not fully capture the structural information compared to graph neural networks (GNNs), which remain the best-performing model type on Matbench. It does go beyond pure compositional understanding, successfully learning and leveraging structural cues to improve performance.

In 2022, in addition to Matbench (Property),¹⁸ a new benchmark called Matbench-Discovery^{21,22} was introduced. This benchmark is designed to evaluate ML energy models on a task that closely simulates a high-throughput discovery campaign for new stable inorganic crystals. Although the recently added models (current leaders) are based on Universal Interatomic Potentials (UIPs), which consider energy, force, and stress, energy-based models still provide a baseline for us. In Table 1 (unique WBM dataset³⁵ as test), Table S1 (full WBM dataset³⁵ as test) and Table S2 (10k most stable subset of WBM dataset³⁵), we summarize the performance of AlchemBERT compared to other energy models following the official test procedure. AlchemBERT’s performance, which significantly surpasses that of Dummy and Voronoi RF,³⁶ demonstrates that it effectively learns from the tokens. It achieves a performance level comparable to BOWSR,³⁷ a Bayesian optimization and graph deep learning model. In addition, for datasets containing either full or unique data, describing a material’s structure in natural language produces better results than merely using CIF files. This is consistent with the previous Matbench results.

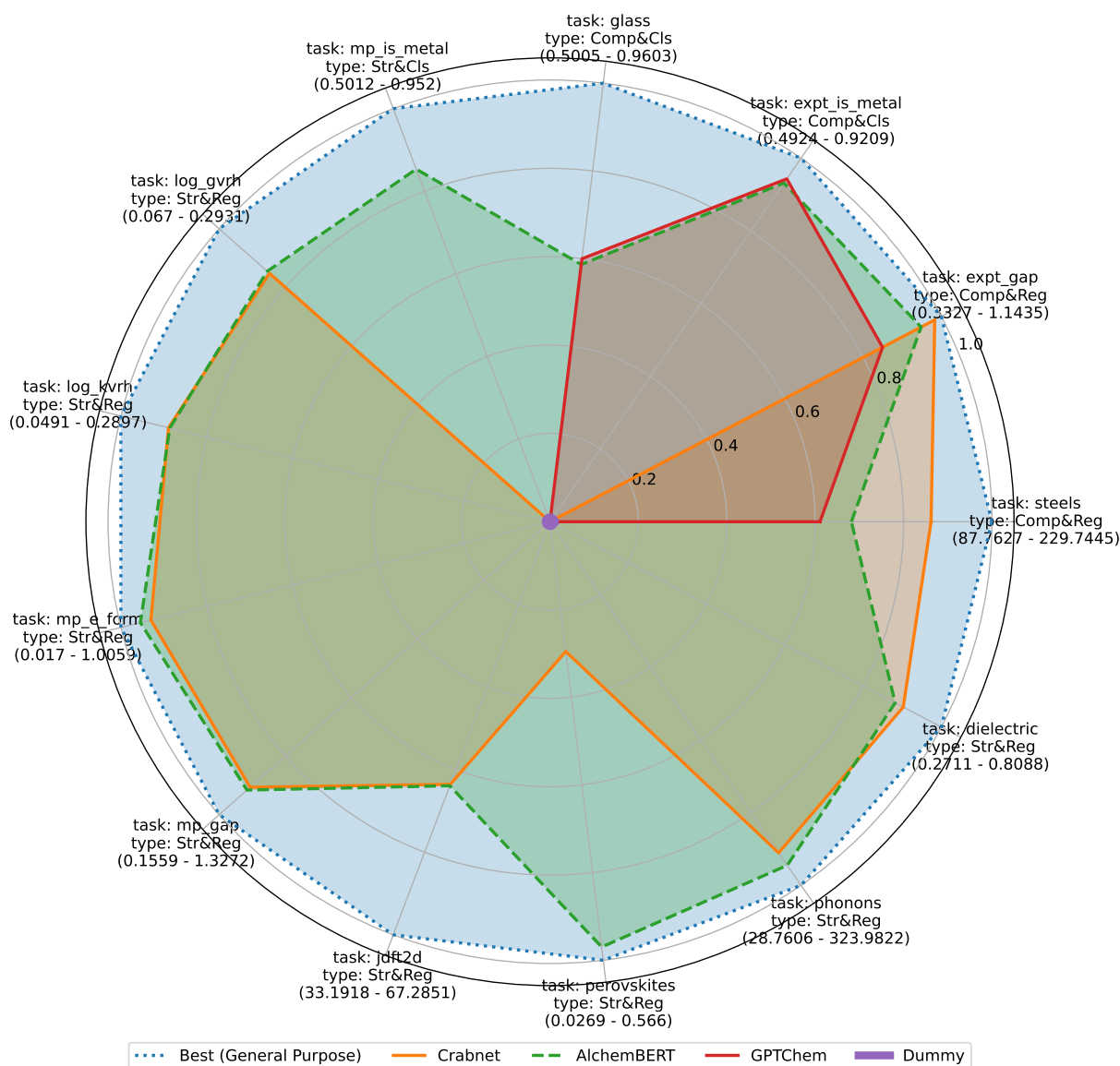


Figure 3: Models comparison between Crabnet,²⁶ GPTChem¹⁹ and ours (AlchemBERT). The task types in the figure are as follows: 'Cls' refers to classification, 'Reg' refers to regression, 'Str' refers to structure input, and 'Comp' refers to composition input.

Table 1: Matbench-Discovery “Unique Prototypes” dataset results for models relying solely on energy.

Model	F1 $\uparrow$	DAF	Precision	Accuracy	MAE	R^2
ALIGNN ³⁸	0.567	3.206	0.490	0.841	0.093	0.297
MEGNet ³⁰	0.510	2.959	0.452	0.826	0.130	-0.248
CGCNN ^{18,28}	0.507	2.855	0.436	0.818	0.138	-0.603
CGCNN+P ³⁹	0.500	2.563	0.392	0.786	0.113	0.019
Wrenformer ⁴⁰	0.466	2.256	0.345	0.745	0.110	-0.018
BOWSR ³⁷	0.423	1.964	0.300	0.712	0.118	0.151
AlchemBERT NL	0.421	2.001	0.306	0.713	0.117	0.096
AlchemBERT CIF	0.419	1.914	0.293	0.684	0.122	0.055
Voronoi RF ³⁶	0.333	1.579	0.241	0.668	0.148	-0.329
Dummy	0.185	1.000	0.154	0.687	0.124	0.000

Discussion

Not only on composition tasks but also on structure-related tasks, the 110-million-parameter AlchemBERT demonstrates strong potential as a property predictor. The application of formation energy prediction on Matbench-Discovery yields competitive results compared to traditional human-designed GNNs. Due to computational power constraints, we have not extensively explored various prompt formats. Nevertheless, even with the current natural language prompts, AlchemBERT demonstrates promising performance compared to raw inputs, where ‘raw’ refers to CIF in structure and formula in composition. Real-world materials often exhibit defects or irregularities, which traditional descriptive methods may struggle to capture due to their complexity. Given the continuous advancements in language models’ ability to understand natural language, using natural language to describe asymmetric or imperfect crystals could offer significant advantages. After all, all current results are obtained on language models that are not particularly large. We hope that future studies will explore novel and innovative prompt designs to further enhance performance and accelerate materials discovery and design.

Methods

BERT Models and Their Training Process

In this study, we utilized the ‘transformers’ library developed by the Hugging Face community⁴¹ together with the PyTorch deep learning framework⁴² to perform our natural language processing tasks.

All models are trained on an Nvidia RTX 4090 with a 9:1 train-validation split. Early stopping is applied with a patience of 200 epochs. The learning rate is set to 1e-5, and a batch size of 32 is used for the pre-trained BERT-base-cased model, which is employed for regression tasks by adding a regression head. Specifically, we append a linear layer to the BERT¹⁵ architecture, which maps the hidden representation of the [CLS] (located at the first position of the sequence) token to a single regression output. Given that BERT-base-cased has a hidden size of 768, the linear layer has dimensions of 768 to 1. The name “BERT-base-cased” refers to the model’s size, with 12 transformer encoder blocks and 110 million parameters, in contrast to the larger BERT-large model (340 million parameters), and is case-sensitive. For composition and structure tasks in Matbench, we use different natural language rules to describe them. We have open source the script and here we can provide some example to illustrate the description methods.

Natural Language Descriptions for Composition and Structure

Unlike the raw input, where we directly input formulas and CIFs generated from pymatgen, we have written a conversion script to transform the composition and structure into natural language. Due to the different input formats, we convert the input data into natural language in three different ways using script.

For Matbench_steel task, for example, in the dataset, one instance input as ‘**Fe0.858 C0.0191 Mn0.00194 Si0.000199 Cr0.0753 Ni0.0000954 Mo0.0341 V0.0101 Nb0.0000603 Co0.000190 Al0.000623**’ is processed as a description:

This is an alloy with the following composition: 85.8% iron (Fe), 7.5% chromium (Cr), 3.4% molybdenum (Mo), 1.9% carbon (C), 1.0% vanadium (V), and smaller amounts of manganese (Mn) (0.1940%), aluminum (Al) (0.0623%), silicon (Si) (0.0199%), cobalt (Co) (0.0190%), nickel (Ni) (0.0095%), niobium (Nb) (0.0060%)

For other composition tasks, an input instance such as **Al0.01Ga0.99P1** is described using:

The chemical formula Al0.01Ga0.99P1 consists of: 0.99 atoms of Ga; 1.0 atoms of P; 0.01 of Al replaced part of Ga in the structure

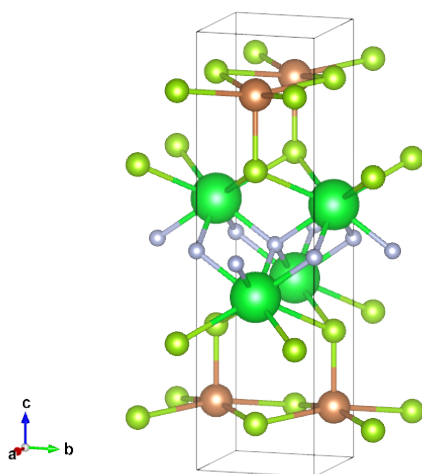


Figure 4: An example (mp-556194 from the Materials Project) described using natural language. In the figure, the large green spheres represent Sr atoms, the small green spheres represent Se atoms, the grey spheres represent F atoms, and the brown spheres represent Sb atoms.

For all structure tasks, an input instance such as the structure shown in Fig.4 is described with:

The chemical formula is SrSbSe₂F. Its crystal structure is in the P4/nmm space group with lattice parameters: a = 4.10 Angstrom, b = 4.10 Angstrom, c = 34.76 Angstrom, alpha = 90.00 degree, beta = 90.00 degree, gamma = 90.00 degree. Detailed atomic interactions include: Sr at Wyckoff position 2c with site symmetry '4mm' has neighboring interactions with: F (2.51 Angstrom away), F (2.51 Angstrom away), F (2.51 Angstrom away), F (2.51 Angstrom away), Se (3.18 Angstrom away), Se (3.18 Angstrom away), Se (3.18 Angstrom away), Se (3.18 Angstrom away); Sb at Wyckoff position 2c with site symmetry '4mm' has neighboring interactions with: Se (2.55 Angstrom away), Se (2.93 Angstrom away), Se (2.93 Angstrom away), Se (2.93 Angstrom away), Se (2.93 Angstrom away); Se at Wyckoff position 2c with site symmetry '4mm' has neighboring interactions with: Sb (2.93 Angstrom away), Sb (2.93 Angstrom away), Sb (2.93 Angstrom away), Sb (2.93 Angstrom away), Sb (2.55 Angstrom away), Sr (3.18 Angstrom away), Sr (3.18 Angstrom away), Sr (3.18 Angstrom away), Sr (3.18 Angstrom away); F at Wyckoff position 2a with site symmetry '4m2' has neighboring interactions with: Sr (2.51 Angstrom away), Sr (2.51 Angstrom away), Sr (2.51 Angstrom away), Sr (2.51 Angstrom away).

For all tasks, we add the prefix '**Please tell me the exfoliation energy (meV/atom) of the following structure:**' before all the previous descriptor to form a complete sentence as model input. And we use the target value from Matbench as it is for the regression. Naturally, 'exfoliation energy' (task name) and 'meV/atom' (task unit) will change depending on the task.

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Supporting Information Available

The source code is available on <https://gitee.com/liuxiaotong15/alchemBERT> for Matbench property tasks and <https://github.com/WangYuHang-WYH/AlchemBERT> for Matbench-Discovery tasks.

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